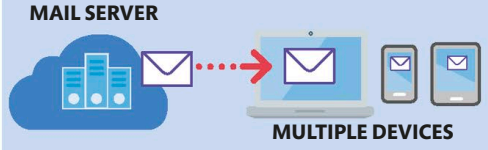
EMAIL RETRIEVAL PROTOCOLS

Emails are sent between computers using SMTP, but to receive emails, the recipient uses an email client that follows either Post Office Protocol (POP) or Internet Message Access Protocol (IMAP). These two sets of rules handle received emails in different ways.

IMAP



- The mail client syncs with server.
- Emails can be accessed and synced across many devices.
- Emails and attachments are not automatically downloaded to a device.
- Original sent and received messages are stored on a server.

POP3



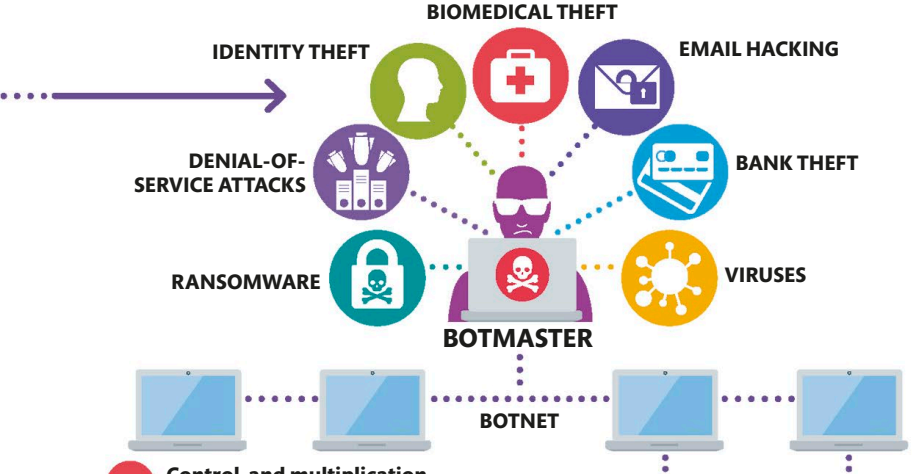
- The mail client and server are not synced.
- Emails can be accessed only from a single device.
- Emails are automatically downloaded to the device then deleted from the server.
- Sent and received messages are stored on the device.



4 Email sent to delivery agent
If the recipient's mail server is found, the message is transferred to his or her mail delivery agent (MDA), using a transmission process described by the SMTP. The message may pass through several MTAs first.

5 Delivery agent passes on email
The MDA performs the final transfer in the process, taking the message from an MTA and sending it to the recipient's local device. It then files it in the user's correct email inbox.

6 Email received
The recipient opens his or her inbox and reads the new email. The way the email is accessed depends on the protocol (see above) adopted by the user's mail client.



3 Control and multiplication
Hackers send commands to the botnet via their C&C server, instructing the computers to perform malicious activities. Meanwhile, the hackers continue adding computers to the botnet.

EMAIL ENCRYPTION

Email encryption prevents emails from being read by anyone other than the intended recipient by using public key encryption. The contents of an encrypted email can be decrypted only by using the correct mathematical key. At its simplest, the sender uses the recipient's public key to encrypt the message, which only the recipient can decrypt with his or her private (secret) key.

